

Vertigo: a diagnosis

1. Key features in the history

(BMJ 2019;366:l5215)



Red Whale

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Guidelines & Evidence Made Simple

Is it vertigo? The sensation of false movement of the world, an internal sensation of movement, tilt or spatial disorientation

Use the **ROTATE** mnemonic to guide your history and gather information (below)

See diagnostic guide: see section 2

Risk factors for CVD

- Increases risk of acute stroke

Other symptoms

Neurological:

- Stroke (acute)
- Cerebellopontine angle tumour (gradual)
- Photophobia, nausea, headache (may suggest vestibular migraine)

Cardiac:

- Chest pain, palpitations or breathlessness (may suggest arrhythmia, angina or aortic stenosis)
- Visual blurring or muffling of sound (may suggest orthostatic hypotension)

Triggers

- Change of head posture (suggests BPPV)
- Dizzy on standing (suggests hypotension)
- Medication, e.g. diuretics cause hypotension; or, if hospitalised recently, consider gentamicin

Age

- Migraine more common in the younger patient
- Stroke more likely with increasing age

Timing of onset

- Sudden onset (seconds) happens with stroke and BPPV, but BPPV tends to settle rapidly
- Migraine or vestibular neuritis start over several seconds to minutes
- Slow, progressive, e.g. cerebellopontine angle tumour (rare: 1 in 100 000/y)

Episodic (is it recurrent?)

- BPPV, vestibular migraine, Ménière's disease
- However, vestibular neuritis (neuronitis) or labyrinthitis tend to happen just once

Vertigo: a diagnosis

2. Possible causes

(BMJ 2019;366:l5215, 2012;345:e5809, BJGP 2020;70:310, BMJ 2022;378:e069850, J Neurol 2009;256:333)

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Is patient describing something other than vertigo?

- Lightheadedness rather than vertigo
- Palpitations, chest pain, dyspnoea
- Loss of consciousness
- May have postural BP drop....

Possible cardiac cause

Cardiac investigations
See article *Transient loss of consciousness*



If vertigo present, consider the following... (treatment and more info in main Vertigo article)

Gradual onset Progressive

- **Associated hearing loss**
- May have facial numbness/weakness, loss of corneal reflex

Cerebellopontine angle tumour

REFER for imaging/2ww

Sudden (immediate) onset

- **Constant vertigo (even if still)**
 - **Older age**
 - **Vascular risk factors**
- May have diplopia, difficulty standing, sensory or motor symptoms
If present >24h = 'acute vestibular syndrome'

Acute stroke (uncommon)
Vestibular neuritis (rare)
Labyrinthitis (even rarer)

ADMIT (needs imaging)

- **Instant onset**, resolves quickly (often triggered by change in head position)
 - **Episodic and recurrent** (previous episodes?)
 - **Normal between attacks**
- May have a positive Dix-Hallpike test

BPPV (commonest cause)

Epley manoeuvre

Rapid onset (minutes)

- **Rapid onset (mins) and lasts mins to days**
 - **Episodic and recurrent** (previous episodes?)
 - Headache may come before, during or after attack, and often not as prominent as in other types of migraine
 - **Look for history of migraine**
- May have nausea/vomiting, headache or photophobia

Vestibular migraine

Consider conventional migraine treatments and antiemetics
See article *Vestibular migraine*

- **Episodic and recurrent**
 - **Hearing loss** (fluctuating, sensorineural)
 - **Vertigo**
 - **Tinnitus**
- Classical triad of Ménière's

Ménière's disease
(beware: often overdiagnosed)

Attacks last >20mins, usually <2h (no more than 24h)
May report fullness in ear
Examination normal between attacks

REFER ENT
Specialist diagnosis

Longstanding Most days for >3m

- Episodic and recurrent
 - Non-spinning vertigo/unsteady feeling
 - Worse with complex visual stimuli
 - Improves with distraction
- May have distress and limitation of activities

Persistent postural-perceptual dizziness (a chronic functional disorder) – see main *Vertigo* article for more detail

Vestibular rehabilitation